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5.2P

## Bag.cs

```
namespace SwinAdventure
{
   public class Bag : Item
   {
      private Inventory _inventory;
      public Bag(string[] ids, string name, string desc) : base(ids, name, desc)
      {
         _inventory = new Inventory();
      }
      public GameObject Locate(string id)
      {

         if (AreYou(id))
         {
            return this;
         }
         return _inventory.Fetch(id);
      }
      public override string FullDescription
      {
         get
         {
```

```
            string itemsDescription = _inventory.ItemList;
            if (string.IsNullOrEmpty(itemsDescription))
            {
                return $"{base.FullDescription}. It's empty.";
            }
            else
            {
                return $"{base.FullDescription}. In the {Name}, you can
see:\n{itemsDescription}";
            }
        }
    }

    public Inventory Inventory
    {
        get { return _inventory; }
    }
  }
}
```

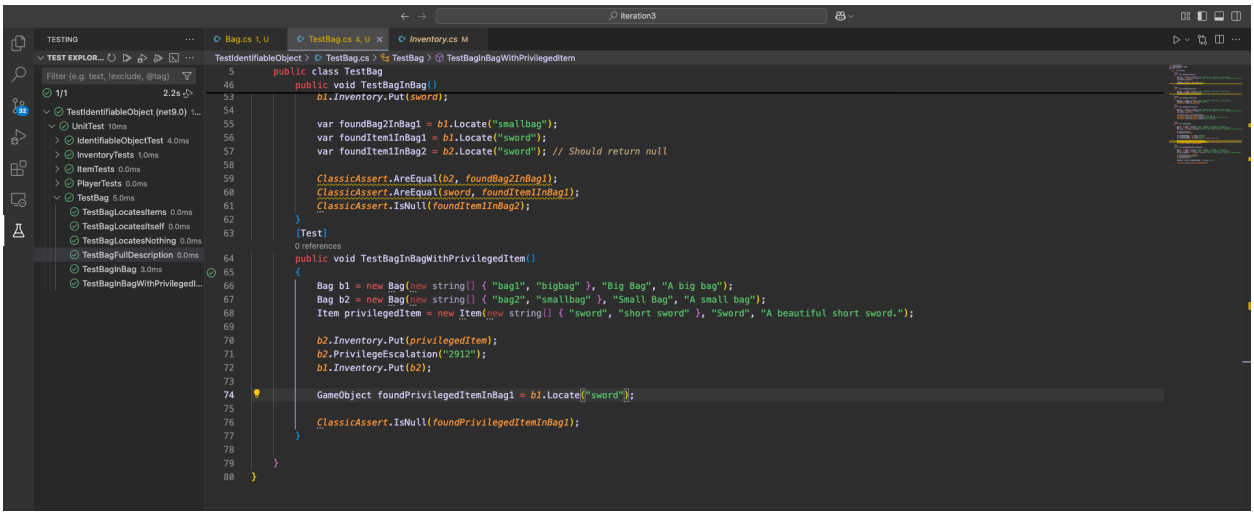
## TestBag.cs

```
using SwinAdventure;
using NUnit.Framework.Legacy;
namespace UnitTest
{
  public class TestBag
  {
```

```
    [Test]
    public void TestBagLocatesItems()
    {
        Bag bag = new Bag(new string[] { "bag", "smallbag" }, "Small Bag", "A small bag");
        Item item1 = new Item(new string[] { "sword", "a short sword" }, "Sword", "A beautiful short sword.");
        bag.Inventory.Put(item1);

        GameObject foundItem = bag.Locate("sword");
        ClassicAssert.AreEqual(item1, foundItem);

    }
    [Test]
    public void TestBagLocatesItself()
    {
        Bag bag = new Bag(new string[] { "bag", "smallbag" }, "Small Bag", "A small bag");
        GameObject foundBag = bag.Locate("bag");
        ClassicAssert.AreEqual(bag, foundBag);
    }
    [Test]
    public void TestBagLocatesNothing()
    {
        Bag bag = new Bag(new string[] { "bag", "smallbag" }, "Small Bag", "A small bag");
        GameObject foundNothing = bag.Locate("bigbag");
        ClassicAssert.IsNull(foundNothing);
    }
```

```
    [Test]
    public void TestBagFullDescription()
    {
        Bag bag = new Bag(new string[] { "bag", "smallbag" }, "Small Bag", "A small bag");
        Item item1 = new Item(new string[] { "sword", "a short sword" }, "Sword", "A beautiful
short sword.");
        bag.Inventory.Put(item1);

        string description = bag.FullDescription;
        ClassicAssert.IsTrue(description.Contains("A small bag."));
        ClassicAssert.IsTrue(description.Contains("In the Small Bag, you can see:\n"));
        ClassicAssert.IsTrue(description.Contains("Sword (sword)"));
    }

    [Test]
    public void TestBagInBag()
    {
        Bag b1 = new Bag(new string[] { "bag1", "bigbag" }, "Big Bag", "A big bag");
        Bag b2 = new Bag(new string[] { "bag2", "smallbag" }, "Small Bag", "A small bag");
        Item sword = new Item(new string[] { "sword", "short sword" }, "Sword", "A beautiful
short sword.");

        b1.Inventory.Put(b2);
        b1.Inventory.Put(sword);

        var foundBag2InBag1 = b1.Locate("smallbag");
        var foundItem1InBag1 = b1.Locate("sword");
```

```
        var foundItem1InBag2 = b2.Locate("sword"); // Should return null


        ClassicAssert.AreEqual(b2, foundBag2InBag1);
        ClassicAssert.AreEqual(sword, foundItem1InBag1);
        ClassicAssert.IsNull(foundItem1InBag2);
    }
    [Test]
    public void TestBagInBagWithPrivilegedItem()
    {
        Bag b1 = new Bag(new string[] { "bag1", "bigbag" }, "Big Bag", "A big bag");
        Bag b2 = new Bag(new string[] { "bag2", "smallbag" }, "Small Bag", "A small bag");
        Item privilegedItem = new Item(new string[] { "sword", "short sword" }, "Sword", "A
beautiful short sword.");


        b2.Inventory.Put(privilegedItem);
        b2.PrivilegeEscalation("2912");
        b1.Inventory.Put(b2);


        GameObject foundPrivilegedItemInBag1 = b1.Locate("sword");
        ClassicAssert.IsNull(foundPrivilegedItemInBag1);
    }
  }
}
```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TEST EXPLOR...
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TestIdentifiableObject (net9.0) 1...
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IdentifiableObjectTest 4.0ms
InventoryTests 1.0ms
ItemTests 0.0ms
PlayerTests 0.0ms
TestBag 5.0ms
TestBagLocatesItems 0.0ms
TestBagLocatesItself 0.0ms
TestBagLocatesNothing 0.0ms
TestBagFullDescription 0.0ms
TestBagInBag 3.0ms
TestBagInBagWithPrivilegedI...
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TestIdentifiableObject > TestBag.cs > TestBag > TestBagInBagWithPrivilegedItem
public class TestBag
public void TestBagInBag()
b1.Inventory.Put(sword);
var foundBag2InBag1 = b1.Locate("smallbag");
var foundItem1InBag1 = b1.Locate("sword");
var foundItem1InBag2 = b2.Locate("sword"); // Should return null
ClassicAssert.AreEqual(b2, foundBag2InBag1);
ClassicAssert.AreEqual(sword, foundItem1InBag1);
ClassicAssert.IsNull(foundItem1InBag2);
}
[Test]
0 references
public void TestBagInBagWithPrivilegedItem()
{
Bag b1 = new Bag(new string[] { "bag1", "bigbag" }, "Big Bag", "A big bag");
Bag b2 = new Bag(new string[] { "bag2", "smallbag" }, "Small Bag", "A small bag");
Item privilegedItem = new Item(new string[] { "sword", "short sword" }, "Sword", "A beautiful short sword.");
b2.Inventory.Put(privilegedItem);
b2.PrivilegeEscalation("2912");
b1.Inventory.Put(b2);
GameObject foundPrivilegedItemInBag1 = b1.Locate("sword");
ClassicAssert.IsNull(foundPrivilegedItemInBag1);
}
}
}